lowThrust — how it works

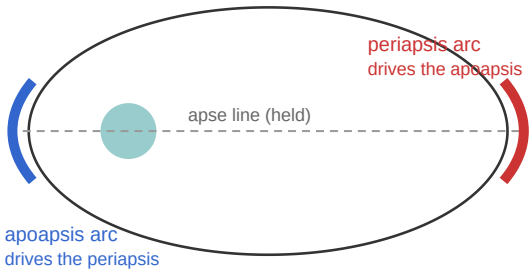
Low-thrust transfers within one SOI, to a moon, to another planet or to a planet's moon: the solver, its phase machine and the stand-in solar system it flies in. No kRPC side yet.

What it does

Given the vessel's current orbit and a target orbit (apoapsis, periapsis, inclination, longitude of the ascending node, about any body; anything left out is free), it decides every tick whether to thrust, which way and how hard, until the target orbit is reached. Every decision comes from the current state and a fresh prediction; there is no stored schedule, so a weak or lost tick is made up later by the same law.

module	role
`orbit.py`	elements of any conic, exact propagation (universal variables), time to periapsis, Lambert, radius toward a fixed direction
`guidance.py`	the within-SOI law: `lock` (apse line, target plane), `command`, arrival test
`plan.py`	flies the law in the simulator to pick the arc width that meets a deadline
`sim.py`	single-body stand-in; the vessel (constant mass, thrust, Isp); the shared RK4 step and tick
`bodies.py`	stock bodies on Keplerian rails with their spheres of influence
`transfer.py`	targeting (closest approach, SOI entry, aim error, correction, finite moon push, window, parking orientation, Lambert re-target, moon hazards) and the capture law
`mission.py`	the phase machine of a whole transfer in patched conics; waiting orbits; routing and leg chaining
`tools/`	`simfly.py` (one SOI, deadline), `simmission.py` (a transfer: timeline, burns, dv), `sweep.sh` (destination × thrust × date grid)

The within-SOI law: Hohmann in many small burns



Two arcs on a fixed apse line; each fixes the opposite apsis.

Decoupled. A tangential burn at periapsis moves the apoapsis and, to first order, not the periapsis, and the other way round. The quantity driven is the radius toward the far end of the apse line, $r = p / (1 + e \cdot u)$, smooth through circular; its rate per m/s of prograde is differenced numerically (`RATE_DV`).

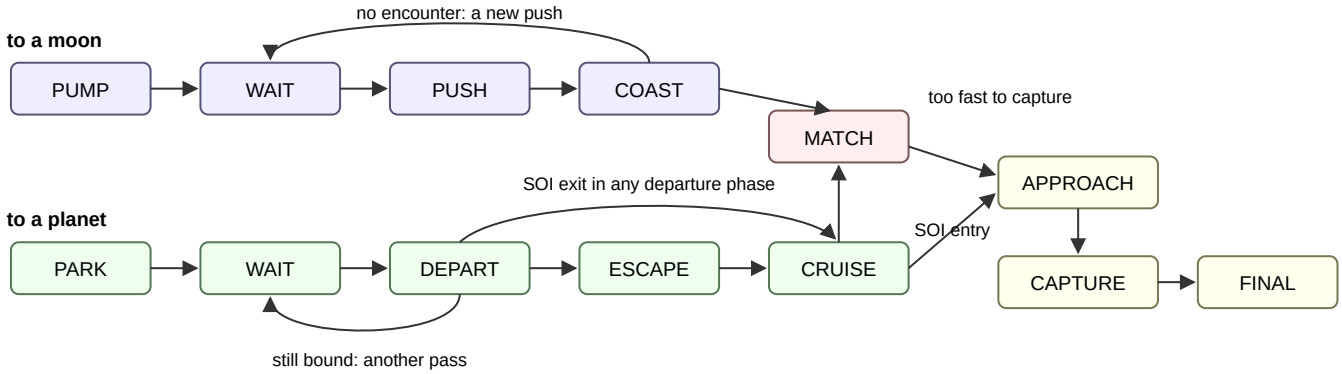
Arcs are ±acos(η) wide (default η 0.99, ±8°). A burn starts half its remaining duration before the apsis and runs at full throttle until the radius is on target or the arc ends. Prograde or retrograde only, so plain SAS modes hold it. **Order**: outward first when raising, inward first when lowering. **Apse line**: held through each burn, re-locked to the osculating periapsis while coasting; a held line (the parking ellipse) is corrected at quadrature once the apsides are close.

Plane change. The target plane is fixed once (`lock`). Normal/anti-normal arcs of the same width at the two nodes, burn time $\delta \cdot h / (r_{node} \cdot a)$; a node below η of the other node's radius is skipped. The plane waits until the apoapsis is raised and the apoapsis is not lowered until the plane is done, so the plane changes on the biggest orbit. Where a node arc and an apsis arc overlap the two burns are one vector, in proportion to the seconds each still needs.

Deadline. `plan.plan` flies the law at `PLAN_GRID` (13) arc widths between η 0.999 and 0.2 and takes the narrowest (cheapest) that arrives within the budget.

Arc width in a mission. Orbit-shaping phases widen the arcs (cutoff down to `ETA_WIDE_MIN` 0.5, ±60°) until the apsis changes still to make fit `ARC_PASSES_MAX` (30) passes; while the orbit crosses another moon's band (within its SOI's reach) they are the widest at once, for the fewest revolutions there.

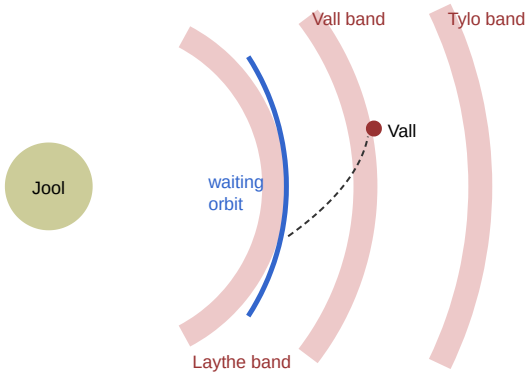
Transfers between bodies: the phase machine



The phases (`mission.Mission`), each re-decided every tick. TRIM (a phasing burn), the floor guard and an evasive burn in an unplanned SOI can interrupt them. A planet's moon is two legs: the planet leg ends in the moon leg's waiting orbit.

phase	what it does	ends when
PUMP	within-SOI law to the waiting orbit (below), in the moon's plane	reached
WAIT (moon)	at each push point, ahead of it: solve the finite prograde push centred there that puts a direct encounter (within 0.75 revolution) on the aim. Skip it if its coast meets another moon, or if it arrives faster than the capture takes (the bar rises 25% a skip, at most once a synodic period). A decided push locks its point.	half the push before its point
PUSH	prograde on a ledger (planned dv less delivered), refined by finite-burn re-solves round the ledger	ledger spent
COAST / CRUISE	corrections onto the aim when one needs more than 0.5 m/s, rechecked every tenth of the time left. To a moon, no encounter (beyond 1.5 SOI or a correction over 150 m/s) goes back to WAIT. To a planet, a miss beyond min(20 SOI, 1% of the target's distance) or a correction over 150 m/s is re-planned as a Lambert arc; the encounter is looked for round the planned arrival. Starts MATCH when planned.	SOI entry
PARK	within-SOI law, held apse line, to the parking ellipse oriented on the window's departure asymptote	reached
WAIT (planet)	coast to the periapsis pass nearest the window; skip a pass whose departure burn, flown forward on RK4 as DEPART flies it, meets a moon	half the push before periapsis
DEPART	prograde until the hyperbolic excess is the window's, or 60° past periapsis (signed anomaly)	escaping: ESCAPE; bound: WAIT
ESCAPE	coast out	SOI exit → CRUISE
MATCH	brakes about the star (a moon: about the planet) onto an aim point just outside the SOI (below)	600 s before the aim point, or at twice the speed it set out to shed
APPROACH	dodges a moon on the inbound leg (below); nudges the periapsis onto the capture altitude along its gradient; coasts an unplanned flyby	the capture law says burn
CAPTURE	the capture law (below); coasts through another body's SOI and re-decides on exit	bound → FINAL; lost → fails
FINAL	within-SOI law to the target orbit	arrived

Moon transfers: waiting orbit, push, aim

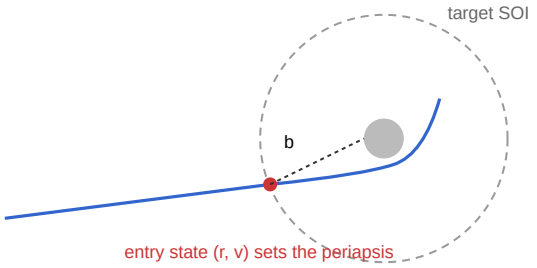


Arriving from above, a weak engine waits in the highest gap between other moons' bands under the target (Vall: between Laythe's band and its own), so the push crosses nothing and arrives slowly.

Waiting orbit (`waiting_orbit`): apoapsis 0.6 of the way to the moon; periapsis as low as possible but high enough that the engine captures the arrival — in one pass (at the lowest capture periapsis that allows), or by the long capture inside the SOI up to its limit (`moon_vcap_long`: the long capture's fastest, at least a quarter of the escape speed at the SOI edge) — and never inside another moon's band.

From above (a planet leg ends in this orbit): the whole orbit in the highest gap between other moons' bands under the moon, at the lowest working periapsis in it. From below with a moon inside the target's: the whole orbit under that moon, its periapsis raised until the push across fits one pass. Nothing capturable: the highest periapsis clear of the others (the slowest arrival; MATCH sheds the rest).

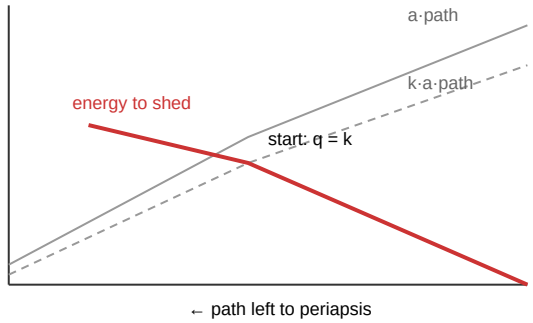
Push points: the periapsis, and every 30° where a push costs no more than `PUSH_OFF_PERIAPSIS` (1.1) of the periapsis Hohmann — those decided far enough ahead to start the longest such push half its burn early. The push is the finite burn the flight flies (`prograde_burn`: RK4 along the velocity, centred on the point when planning, from now when re-solving), searched up to `PUSH_DV_FACTOR` (2) times the tangential push that reaches the moon's orbit.



The aim is measured where the patched conic is decided: at the SOI entry.

Aim (`transfer.aim_error`): when the coast enters the target's SOI, the entry state relative to the target (speed w, energy $E = w^2/2 - \mu/R_{SOI}$) gives the miss distance across the velocity that makes periapsis r_c: $b = \sqrt{(2E\, r_c^2 + 2\mu\, r_c)} / w$, at most 0.95 of the SOI (a slow entry is bound on arrival; APPROACH lifts the rest). When the coast does not enter, the closest approach against the straight-line $b = r_c\sqrt{(1 + 2\mu/(r_c v^2) - 2\mu/(R_{SOI} v^2))}$, at most 0.8 SOI. The aim lies on the side prograde about the target's orbit normal — or over a pole (below). The push, the mid-course correction (numerical Jacobian of the two miss components against the three impulse axes, minimum-norm) and MATCH share it.

The capture law



q = energy to shed / (a × path left to periapsis): coast until q reaches k, then throttle at q.

The capture only has to make the orbit **bound**: below 0.3 of the SOI (or the final apoapsis), or once past periapsis below the highest apoapsis clear of the planet's moons — always with the periapsis above the floor (30 km over the atmosphere). FINAL shapes the orbit.

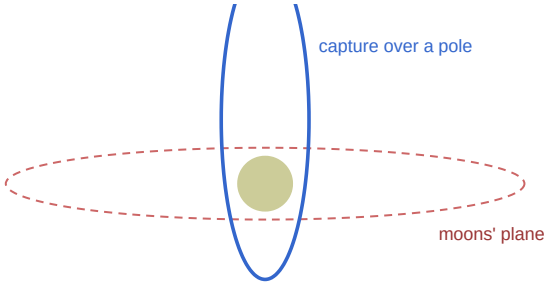
Short: if what is left fits in 80% of the time within ±60° of periapsis, the burn is centred on periapsis at full throttle. **Long**: thrust along the velocity removes specific energy at exactly a per metre of path, wherever it is flown; the law coasts until the energy above a bound orbit reaches `CAPTURE_PATH_SHARE` (k 0.8) of what full thrust removes on the path left to periapsis, then throttles at that ratio. Outbound and unbound: full throttle.

Direction: retrograde (sized to one tick's dv), less its part along the periapsis gradient once the periapsis is at or below the aim (energy shed without lowering it); a gradient term holds the periapsis between the floor and 1.5× the aim; below the floor, only the lift. Steps change the velocity by at most 5%. It runs only inside the SOI. Planning flies the same law (`fly_capture`) to choose between a capture and MATCH.

MATCH: braking before the SOI

When: planned when the arrival is faster than a capture starting within 0.8 of the SOI can take; its arrival speed v_s is 0.8 (`ARRIVE_MARGIN`) of the fastest the capture takes. A moon leg plans it too, in the planet's frame. **Start**: from the coast's predicted encounter, when braking from v to v_s at `MATCH_SHARE` (0.8) of the engine, plus the standoff, needs all the time left. **Law**: zero-effort-miss / zero-effort-velocity guidance, a = 6 ZEM/t² − 2 ZEV/t, ZEM and ZEV from coasting the vessel and the destination t ahead, $t = 2(v \cdot t_{ca} - d_{back})/(v + v_s)$. **Aim point**: `MATCH_STANDOFF` (1.05) SOI radii out along the arrival direction, moving at v_s, offset by the aim's miss distance. **End**: 600 s before the aim point, once $v \le 1.05\ v_s$, or once it has spent twice the speed it set out to shed.

Planets with moons: capture plane, descent, hazards



Free target plane at a planet with moons: the aim point is over a pole, and the descent crosses each moon's radius far out of its plane.

Polar capture: with moons and the target plane left free, the aim lies in the plane through the poles (correction, Lambert re-target and MATCH alike).

Deep wells (Jool): a tight capture (apoapsis under the innermost moon) is compared, by predicted dv, with a loose one (0.5 SOI) plus the impulsive cost of lowering it afterwards.

Approach: every tenth of the time left, if the inbound coast, flown through any moon's SOI in patched conics, would hit the moon, leave the system, or come out with the periapsis under the floor, the cheapest burn across the velocity (8 directions, doubling sizes, the periapsis put back on the aim by two Newton steps along its gradient; cost = the burn plus the plane change it adds against a fixed target plane, at the capture orbit's speed) that keeps the periapsis in the capture's band and clears the moons, flown as a ledger. A harmless flyby is left alone.

Arc burns: on its first tick, each orbit-shaping burn is flown in prediction by the guidance law itself, closed loop (`_fly_arc`), and capped at the largest tenth of it whose next revolution stays out of every other moon's SOI; if none does, the one that meets it latest. The cap is delivered exactly, from the first tick. **Phasing trims**, at most once per half revolution: if the rest of this revolution meets a moon, a trim now; else, if the next one does, a trim at the next periapsis — the smallest along-track burn, of ±0.5–8 m/s and of a quarter to twice the burn that slips the pass past a moon's SOI (dP/dv = 3aVP/μ; at least 0.1 m/s), that clears the coming revolutions. **Evasive burn**: inside an unplanned SOI with the periapsis below the floor, along the periapsis gradient. **Floor guard**: outside CAPTURE and DEPART, a bound periapsis inside the atmosphere or surface is lifted first. After a slingshot, braking until the apoapsis is inside 0.6 SOI.

Departure and the window

Window: Lambert over departure dates (one synodic period from the pump's estimated end) and flight times (0.5–1.5 of the Hohmann's), least |v∞ out| + |v∞ in|. **Parking ellipse**: periapsis above the atmosphere, apoapsis 0.9 of the highest clear of the planet's moons, its plane containing the departure asymptote (the cheapest to reach). A

weak engine's departure takes several periapsis passes; whatever v∞ it leaves with, the cruise's re-target builds the rest about the star.

The stand-in

Patched conics: the vessel's state is relative to the body whose SOI it is in. A coasting tick is propagated exactly (universal variables) and bisected to an SOI boundary when it crosses one; the frame then switches by the body's state on its rails. Thrusting ticks are `THRUST_SUBSTEPS` (4) RK4 steps and change the velocity by at most `THRUST_DV_SHARE` (2%). Ticks are uniform in angle (r²/h per 2°, capped at period/180). The vessel has constant mass and unlimited propellant: dv is thrust/mass × burn time, and xenon is reported from it with the rocket equation. A flight ends on arrival, impact, the time limit (`MISSION_MAX_DAYS`) or a failed capture.